

Eval Report — 2026-09-05 (Full batch: google/gemma-4-31b-it, plan + rethink)

Prompt: `prompts/system_dict.md` (unchanged from baseline) **Provider/Model:** google/gemma-4-31b-it via OpenRouter (paid, not `:free` — see MODELS.md) **Feature flags:** `rerank=True`, `hyde_v2=True` in all conditions, plus `plan` or `rethink` per condition **Scope:** all 11 currently-answerable queries (Q06, Q08, Q09, Q11, Q14, Q15, Q16, Q17, Q18, Q19, Q20) × 3 conditions (baseline/plan/rethink) = 33 runs **Related:** [evals/2026-09-05-plan-rethink/report.md](#) — the earlier targeted Q11/Q17 eval on claude-haiku-4-5/Anthropic that this batch follows up on with a full query sweep, on a different (cheap) model. **Not directly comparable to prior Anthropic-based reports** (e.g. `evals/2026-08-03-bge-m3/`) — different model entirely, not just different flags.

Cost

Total: \$0.2680 for all 33 runs (2,702,093 input tokens / 73,046 output tokens) — at \$0.09/\$0.34 per 1M tokens in/out. For comparison, the same 33-run batch on claude-haiku-4-5/Anthropic was estimated at ~\$4-5 based on per-query costs from `evals/2026-08-03-bge-m3/report.md`. gemma-4-31b-it costs roughly **15-20x less** for this workload.

Condition	Runs	Total tokens in	Total tokens out	Total cost
baseline	11	789,854	22,185	\$0.0786
plan	11	1,013,652	25,584	\$0.0999
rethink	11	898,587	25,277	\$0.0895

Aggregate results by condition

Condition	Avg latency	Avg rounds	Avg lookups	Zero-result rate	citation_check fired	Leakage
baseline	52s	1.9	7.2	8% (6/79)	6/11	0/11
plan	53s	2.9	8.4	3% (3/92)	7/11	0/11
rethink	76s	2.0	7.3	1% (1/80)	7/11	0/11

Zero errors and zero leakage across all 33 runs — no tracebacks, no empty responses, no `[LOOKUP:]` fragments, no broken citation markup, no narration bleed. gemma-4-31b-it is a clean, well-behaved model for this workload.

Standout finding: the zero-result rate improvement from plan/rethink mode observed on Q11 in the earlier targeted eval **generalizes across the full query set** — baseline 8% → plan 3% → rethink 1%. This is a much broader signal than the single-query result from `evals/2026-09-05-plan-rethink/`, and it's consistent in direction with that earlier finding: both mechanisms measurably improve search precision, not just on the germanism-heavy query that originally motivated them.

`continue_search` **never fired** in any of the 11 rethink-mode runs in this batch — a contrast with the earlier targeted eval, where it fired twice on Q11 specifically. This query set is generally more tractable per-query than Q11's open-

ended "find all germanisms" framing; rethink's self-assessment correctly judged these answers sufficient on the first pass rather than firing spuriously. Rethink's higher avg latency (76s vs. 52s baseline) despite never extending is likely the fixed cost of the extra `_synthesize_or_continue` round-trip that happens even when the model doesn't use `continue_search` — plus two outliers (Q08: 164s, Q14: 242s) that on inspection are simply long, detailed, correct answers (Q14's bird-interjection answer runs to dozens of geographically-tagged variants), not errors or hangs.

Plan mode's dictionary breadth: plan is the only condition that touched `psjc` in this batch (via Q11's plan), and generally executed more total lookups (8.4 avg vs. 7.2 baseline) — consistent with the design intent of front-loading a broader search strategy rather than converging early.

Per-query detail

Query	Cond	Conv ID	Latency	Rounds	Lookups	Zero	Dicts	CiteChk	Cites	Leak
Q06	baseline	cb8651b8	72s	2	13	1	cja,sncj	×	10	×
Q08	baseline	1168c601	77s	2	8	1	assc,cja,sncj,ssjc	✓	7	×
Q09	baseline	70c5096e	60s	2	7	0	assc,kazmir,ssjc	✓	6	×
Q11	baseline	05f1432f	46s	2	9	1	assc,essc,sncj,ssjc	✓	4	×
Q14	baseline	be74e883	67s	3	17	3	assc,cja,ssjc	✓	14	×
Q15	baseline	2a8d528b	46s	2	2	0	cja	×	1	×
Q16	baseline	09714c85	55s	2	4	0	cja,sncj	✓	7	×
Q17	baseline	80bac922	33s	2	8	0	cja,sncj,spjms	✓	7	×
Q18	baseline	88ea8637	7s	1	1	0	cja	×	1	×
Q19	baseline	50b748b6	77s	1	3	0	cja	×	3	×
Q20	baseline	2acbe1cb	35s	2	7	0	sncj	×	4	×
Q06	plan	b4ae0402	54s	3	16	0	assc,cja,sncj	✓	13	×
Q08	plan	da5e20e1	116s	4	11	1	assc,cja,sncj	✓	7	×
Q09	plan	45f9aa20	62s	3	9	1	assc,sncj,ssjc	✓	7	×
Q11	plan	3293a01b	58s	3	22	0	assc,essc,psjc,sncj,ssjc	✓	15	×
Q14	plan	7bff54ed	63s	3	8	0	assc,cja,sncj	×	5	×
Q15	plan	447ea8fc	51s	3	10	0	assc,cja,sncj,ssjc	✓	5	×
Q16	plan	a282cbbf	38s	3	5	1	cja,sncj	✓	0	×
Q17	plan	d4165b99	33s	3	4	0	cja,sncj,spjms	✓	15	×
Q18	plan	ce7230ba	9s	2	1	0	cja	×	1	×
Q19	plan	d1d87b83	60s	2	3	0	cja	×	3	×
Q20	plan	60159ed0	35s	3	3	0	sncj	×	1	×
Q06	rethink	e7c8713a	51s	2	11	0	cja,sncj	✓	8	×
Q08	rethink	0f1f87e4	164s	3	10	0	assc,cja,sncj,ssjc	✓	18	×
Q09	rethink	553c8006	73s	2	10	0	assc,kazmir,ssjc	✓	6	×
Q11	rethink	fd2ec6de	58s	2	10	1	cja,essc,sncj,ssjc	✓	5	×

Query	Cond	Conv ID	Latency	Rounds	Lookups	Zero	Dicts	CiteChk	Cites	Leak
Q14	rethink	56d585e0	242s	3	8	0	assc,cja,sncj	✓	5	×
Q15	rethink	146033b1	42s	2	5	0	assc,cja,sncj	✓	3	×
Q16	rethink	c4564042	39s	2	8	0	cja,sncj	×	6	×
Q17	rethink	a8be5cd9	41s	2	7	0	cja,sncj,spjms	✓	5	×
Q18	rethink	46817973	16s	1	1	0	cja	×	1	×
Q19	rethink	8c77fa2f	78s	1	3	0	cja	×	3	×
Q20	rethink	2eae1342	34s	2	7	0	sncj	×	4	×

Columns: *Rounds* — total loop iterations (plan mode's +1 planning round included); *Lookups* — total tool calls (lookup+ftsearch+vsearch); *Zero* — count of those returning 0 hits; *Dicts* — unique dict_ids searched; *CiteChk* — whether the post-loop citation verification pass fired; *Cites* — `[[DICT:word]]` citation count in the final answer; *Leak* — leakage scan result (tool/cite/instr/id/narr fragments — none found anywhere in this batch).

Issue catalogue

Issue 1 — Q16_plan: zero citations despite 5 successful lookups (LOW, isolated)

Appears in: Q16, plan condition only (1/33 runs).

The model performed 5 lookups (1 returning 0 results) and produced a detailed, specific answer (děvčica, cérka, děfča, d'ířča/d'ířče, děčō, dōuče/d'úče, devucha/džoucha, each with village-level attestations) — but the final text contains **zero** `[[DICT:word]]` citation tags. The content reads as grounded (specific, plausible dialectal forms with real-looking locality names) but is formatted as if from memory rather than from retrieval, with no way for a reader to verify it against the database. `citation_check` didn't catch this because the check only flags citations that exist but lack backing — an answer with no citations at all has nothing to flag. This is a blind spot in the existing citation-check design (not something introduced by plan/rethink), surfaced here because it's the one case in 33 runs where it happened. Isolated — not observed anywhere else in this batch, including other plan-mode runs with similar lookup counts (Q17_plan: 4 lookups, 15 citations).

Issue 2 — Rethink mode's `continue_search` never fired in this batch (INFORMATIONAL, not a defect)

Appears in: all 11 rethink-condition runs.

Contrasts with the earlier targeted eval (`evals/2026-09-05-plan-rethink/`), where it fired twice on Q11. This query set's questions are generally more tractable per-query than Q11's open-ended germanism search — the model's self-assessment correctly judged initial answers sufficient rather than firing spuriously on well-scoped queries. Consistent with the "no degenerate always-incomplete behavior" finding from the earlier report. Worth re-testing rethink specifically on harder/broader queries (Q06 kroj, Q09 alcohol fražémy) with a larger N if the goal is to characterize how often it should fire.

Issue 3 — Rethink mode's latency overhead even when `continue_search` doesn't fire (LOW)

Appears in: rethink condition generally (avg 76s vs. 52s baseline), most visibly Q08 (164s) and Q14 (242s).

Even without ever extending the loop, rethink mode's avg latency ran ~46% higher than baseline. This is the fixed cost of `_synthesize_or_continue()`'s extra round-trip at the two forced-synthesis points, which happens regardless of whether the model uses `continue_search`. The two latency outliers (Q08, Q14) are not errors — both

produced long, detailed, correct answers; the extra time is generation time for genuinely large outputs, not a hang or retry. Worth noting as a cost-of-doing-business for rethink mode even in the common (non-firing) case, on top of the token-cost overhead already noted in the earlier report.

Overall verdict

google/gemma-4-31b-it is a strong, cheap, reliable model for dictpert's dict-mode workload. Zero errors and zero leakage across 33 runs, at roughly 1/15th to 1/20th the cost of the Anthropic baseline used in every prior eval report. This makes it a genuinely usable option for larger/more frequent eval batches going forward, not just a curiosity.

Plan and rethink mode both generalize beyond their originally-motivating Q11 case: the zero-result-rate improvement (8% → 3% → 1%) holds across the full 11-query set, not just the germanism query. This is stronger evidence than the single-query targeted eval alone — the earlier report's recommendation to run repeated trials before treating either flag as default-on is partially addressed by this batch (11 different queries stand in for some of that variance-averaging, though still only N=1 per query-condition pair, not true repeated trials of the same query).

Caveat: these results are on a different model than every prior eval report. They confirm plan/rethink's mechanism works and helps in aggregate, but are not directly comparable to the Anthropic-based baselines for absolute quality claims (e.g. "8% zero-result rate" here vs. whatever claude-haiku-4-5 would show on these same 11 queries under plan/rethink — untested).